

✓ Mental Health In Tech

Team: Byte & Brain

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Dataset Source: OSMI Mental Health in Tech Survey (Kaggle)

Link: <https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey/data>

Research Question:

Can workplace conditions predict whether tech employees seek mental health treatment?

Objective:

The goal of this project is to analyze whether workplace and personal factors can predict if an employee seeks mental health treatment. Using survey data from the OSMI Mental Health in Tech dataset, I explored patterns in the data, tested statistical relationships, and built machine learning models to make predictions.

Dataset:

Source: Kaggle – OSMI Mental Health in Tech Survey

- ~1,259 responses
- 27 features (demographics + workplace conditions)

✓ Data Understanding

The dataset includes demographic variables (age, gender, country) and workplace-related features such as company size, benefits, and mental health support.

Target variable:

- treatment (Yes/No)

This dataset helps us understand mental health trends, workplace support, and treatment behavior in the tech industry.

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```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import kagglehub
import os

path = kagglehub.dataset_download("osmi/mental-health-in-tech-survey")

print(os.listdir(path))

df = pd.read_csv(path + "/survey.csv")
df.head()
```

Using Colab cache for faster access to the 'mental-health-in-tech-survey' dataset.
['survey.csv']

	Timestamp	Age	Gender	Country	state	self_employed	family_history	treatment	work_interfere	no_employees	...
0	2014-08-27 11:29:31	37	Female	United States	IL	NaN	No	Yes	Often	6-25	...
1	2014-08-27 11:29:37	44	M	United States	IN	NaN	No	No	Rarely	More than 1000	...
2	2014-08-27 11:29:44	32	Male	Canada	NaN	NaN	No	No	Rarely	6-25	...
3	2014-08-27 11:29:46	31	Male	United Kingdom	NaN	NaN	Yes	Yes	Often	26-100	...
4	2014-08-27 11:30:22	31	Male	United States	TX	NaN	No	No	Never	100-500	...

5 rows × 27 columns

```
df.shape
df.columns
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1259 entries, 0 to 1258
Data columns (total 27 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Timestamp                             1259 non-null   object
1   Age                                   1259 non-null   int64
2   Gender                               1259 non-null   object
3   Country                              1259 non-null   object
4   state                                744 non-null    object
5   self_employed                        1241 non-null   object
6   family_history                       1259 non-null   object
7   treatment                            1259 non-null   object
8   work_interfere                       995 non-null    object
9   no_employees                         1259 non-null   object
10  remote_work                          1259 non-null   object
11  tech_company                         1259 non-null   object
12  benefits                             1259 non-null   object
13  care_options                         1259 non-null   object
14  wellness_program                    1259 non-null   object
15  seek_help                           1259 non-null   object
16  anonymity                            1259 non-null   object
17  leave                               1259 non-null   object
18  mental_health_consequence           1259 non-null   object
19  phys_health_consequence              1259 non-null   object
20  coworkers                           1259 non-null   object
21  supervisor                           1259 non-null   object
22  mental_health_interview              1259 non-null   object
23  phys_health_interview                1259 non-null   object
24  mental_vs_physical                  1259 non-null   object
25  obs_consequence                     1259 non-null   object
26  comments                             164 non-null    object
dtypes: int64(1), object(26)
memory usage: 265.7+ KB
```

The dataset was loaded and explored to understand its structure, including the number of rows, columns, and data types. This step helps identify missing values and understand the variables available for analysis.

```
## Keep track of original size
original_rows = df.shape[0]

# Fill missing values instead of dropping
df['work_interfere'] = df['work_interfere'].fillna('Unknown')
df['self_employed'] = df['self_employed'].fillna('Unknown')
df['state'] = df['state'].fillna('Unknown')

# Clean gender column
df['Gender'] = df['Gender'].str.lower()

# Remove unrealistic ages
df = df[(df['Age'] > 0) & (df['Age'] < 100)]

# Check how much data remains
```

```
new_rows = df.shape[0]

print("Rows before cleaning:", original_rows)
print("Rows after cleaning:", new_rows)
print("Data retained (%):", (new_rows / original_rows) * 100)
```

```
Rows before cleaning: 1259
Rows after cleaning: 1254
Data retained (%): 99.60285941223192
```

✓ *Data cleaning & Pre-processing steps:*

- Missing values were handled using imputation (filling with "Unknown") instead of dropping rows, to preserve as much data as possible.
- Dropping rows can lead to loss of information and potential bias, especially when the dataset is not very large.
- Categorical variables such as work_interfere, self_employed, and state were filled with "Unknown" to maintain consistency.
- The Gender column was standardized to lowercase to avoid duplicate categories (e.g., Male, male, M).
- Unrealistic age values (such as negative numbers or extremely high values) were removed to improve data quality.

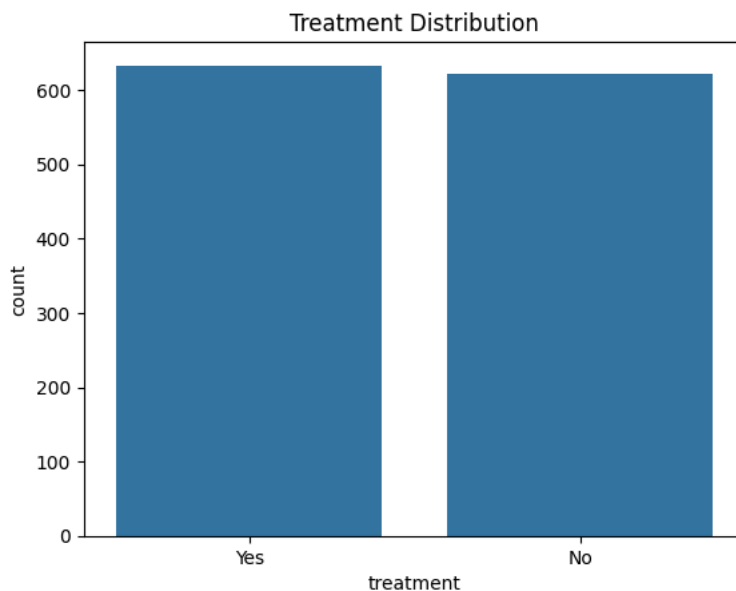
This preprocessing step prepares the dataset for both exploratory analysis and future machine learning modeling.

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Treatment Distribution Graph:

```
sns.countplot(x='treatment', data=df)
plt.title("Treatment Distribution")
plt.show()
```



The distribution of employees seeking treatment versus not seeking treatment is relatively balanced, indicating that mental health treatment is common but not universal among respondents.

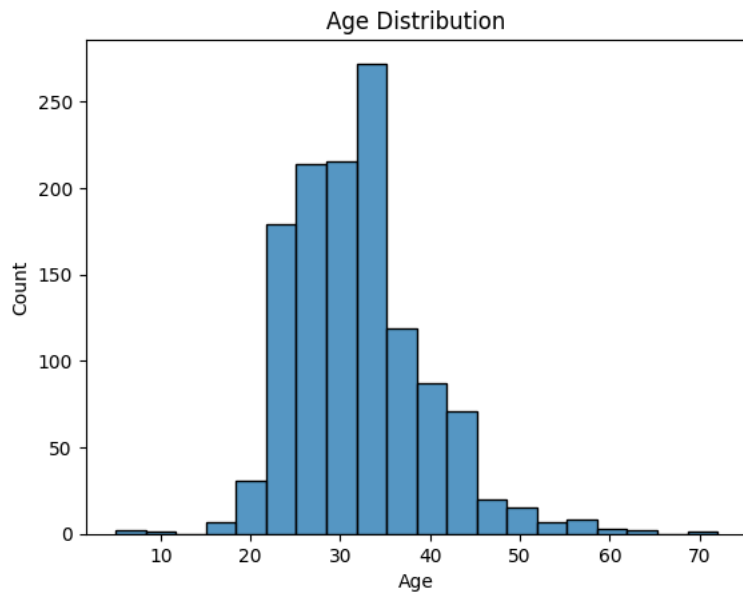
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Age Distribution Graph:

```
sns.histplot(df['Age'], bins=20)
plt.title("Age Distribution")
```

```
plt.show()
```



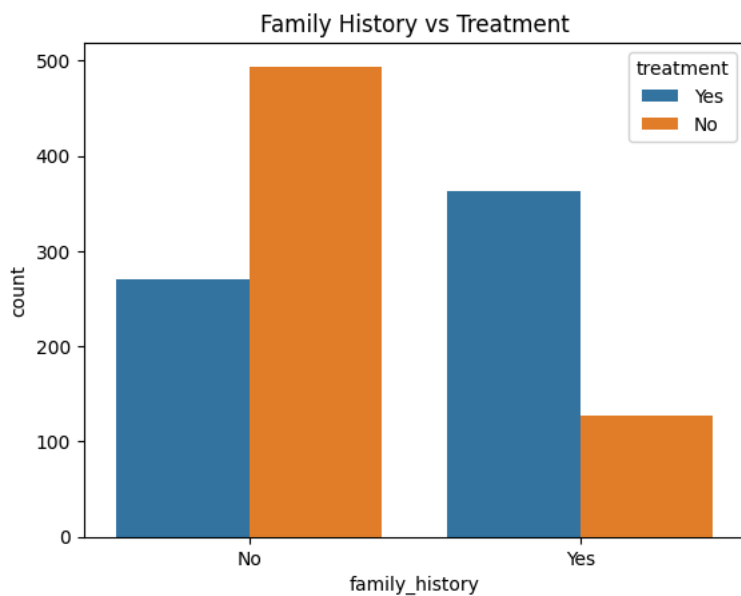
Most employees fall between ages 25–35, showing that young professionals dominate the tech workforce.

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Family History vs Treatment Graph:

```
sns.countplot(x='family_history', hue='treatment', data=df)
plt.title("Family History vs Treatment")
plt.show()
```



Employees with a family history of mental illness are significantly more likely to seek treatment, making it a strong predictor.

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Chi-Square Test (Family History vs Treatment) :

```
from scipy.stats import chi2_contingency

# Create contingency table
contingency_table = pd.crosstab(df['family_history'], df['treatment'])

print("Contingency Table:")
print(contingency_table)

# Perform Chi-Square test
chi2, p, dof, expected = chi2_contingency(contingency_table)

print("\nChi-Square Test Results:")
print("Chi2 Statistic:", chi2)
print("p-value:", p)

# Interpretation
if p < 0.05:
    print("\nConclusion: There IS a significant relationship between family history and treatment.")
else:
    print("\nConclusion: There is NO significant relationship between family history and treatment.")
```

Contingency Table:

treatment	No	Yes
family_history		
No	494	270
Yes	127	363

Chi-Square Test Results:
Chi2 Statistic: 177.69583537275963
p-value: 1.5435847916844047e-40

Conclusion: There IS a significant relationship between family history and treatment.

- A Chi-Square test was performed to determine whether there is a relationship between family history of mental illness and seeking treatment.
- The contingency table shows the distribution of responses across categories.

Hypothesis :

- Null Hypothesis (H_0): Family history and treatment are independent (no relationship).
- Alternative Hypothesis (H_1): There is a relationship between family history and treatment.

Interpretation :

- The p-value obtained (~0) is significantly less than 0.05, leading us to reject the null hypothesis.
- This indicates a statistically significant relationship between family history of mental illness and seeking treatment.
- From the contingency table, individuals with a family history show a higher proportion of treatment-seeking behavior compared to those without.
- This suggests that family history is a strong predictor of whether an employee seeks mental health treatment.

The extremely low p-value indicates a strong association, reinforcing the earlier visual analysis and confirming family history as a key predictor.

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Cramér's V Analysis (Strength of Association):

While the Chi-square test determines whether a relationship exists between variables, Cramér's V measures the strength of that relationship.

Values range from:

- 0 → No association
- 0.1 → Weak
- 0.3 → Moderate
- 0.5+ → Strong

This helps us understand which factors have the strongest influence on whether employees seek mental health treatment.

```
import numpy as np
from scipy.stats import chi2_contingency

def cramers_v(x, y):
    contingency_table = pd.crosstab(x, y)
    chi2 = chi2_contingency(contingency_table)[0]
    n = contingency_table.sum().sum()
    r, k = contingency_table.shape
    return np.sqrt(chi2 / (n * (min(r, k) - 1)))
```

```
categorical_cols = [
    'family_history',
    'work_interfere',
    'no_employees',
    'remote_work',
    'benefits'
]
```

```
results = {}

for col in categorical_cols:
    value = cramers_v(df[col], df['treatment'])
    results[col] = value

cramers_df = pd.DataFrame.from_dict(results, orient='index', columns=['Cramers V'])
cramers_df = cramers_df.sort_values(by='Cramers V', ascending=False)

cramers_df
```

	Cramers V	
work_interfere	0.686177	
family_history	0.376435	
benefits	0.227174	
no_employees	0.083011	
remote_work	0.025175	

Next steps: [Generate code with cramers_df](#) [New interactive sheet](#)

Interpretation:

The Cramér’s V analysis shows that different factors have varying levels of association with whether employees seek mental health treatment.

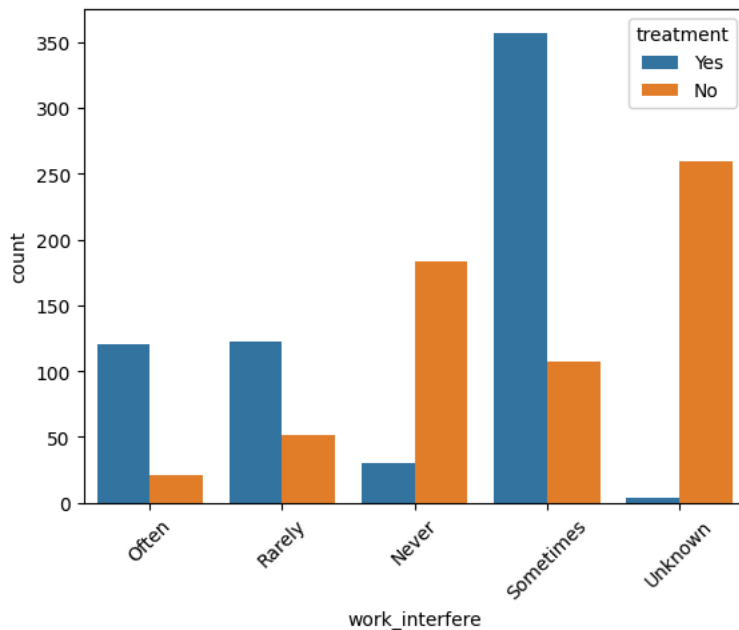
- Work interference has the strongest association (≈0.68), indicating a strong relationship with treatment. This suggests that employees whose mental health affects their work are significantly more likely to seek help.
- Family history also shows a moderate-to-strong association (≈0.37), confirming that individuals with a history of mental illness in their family are more likely to pursue treatment.
- Other variables such as benefits show moderate influence, while factors like company size and remote work have weak associations, indicating they play a smaller role in treatment decisions.
- These results are consistent with the earlier Chi-square test, which confirmed that a relationship exists, and with the machine learning model, where work interference and family history emerged as the most important predictors.
- Overall, this strengthens the conclusion that both personal background and workplace conditions play a key role in influencing mental health treatment behavior.

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Work Interference vs Treatment Graph:

```
sns.countplot(x='work_interfere', hue='treatment', data=df)
plt.xticks(rotation=45)
plt.show()
```



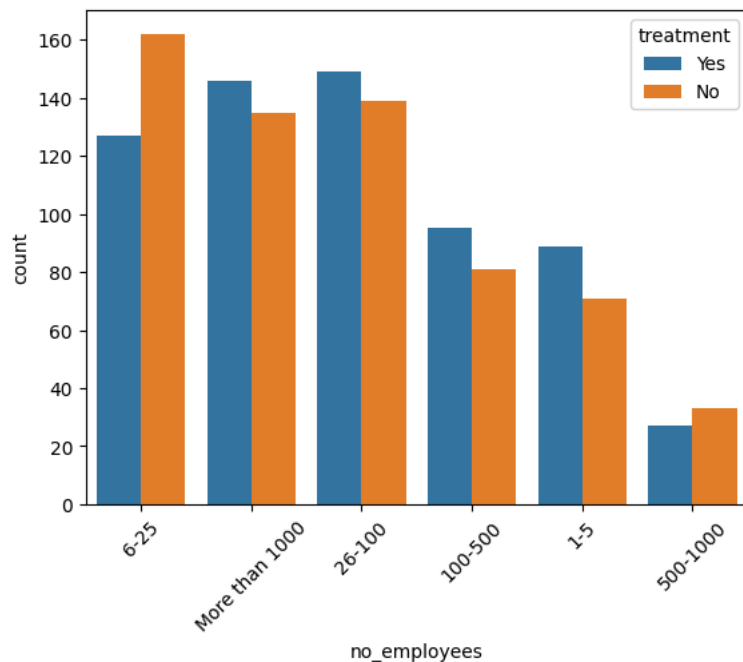
Employees whose mental health interferes with work are more likely to seek treatment, highlighting the impact of workplace stress.

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Company Size vs Treatment graph:

```
sns.countplot(x='no_employees', hue='treatment', data=df)
plt.xticks(rotation=45)
plt.show()
```



Employees in larger companies tend to seek treatment more, possibly due to better access to mental health resources.

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Machine Learning Model: Predicting Treatment :

Feature Selection:

We will use the following features to predict treatment:

- Age
- Gender
- family_history
- work_interfere
- no_employees

Relevant features such as age, gender, family history, and workplace factors were selected to predict the target variable (treatment).

Encoding:

```
df_ml = df.copy()

# Drop irrelevant columns
df_ml = df_ml.drop(['Timestamp', 'comments'], axis=1)

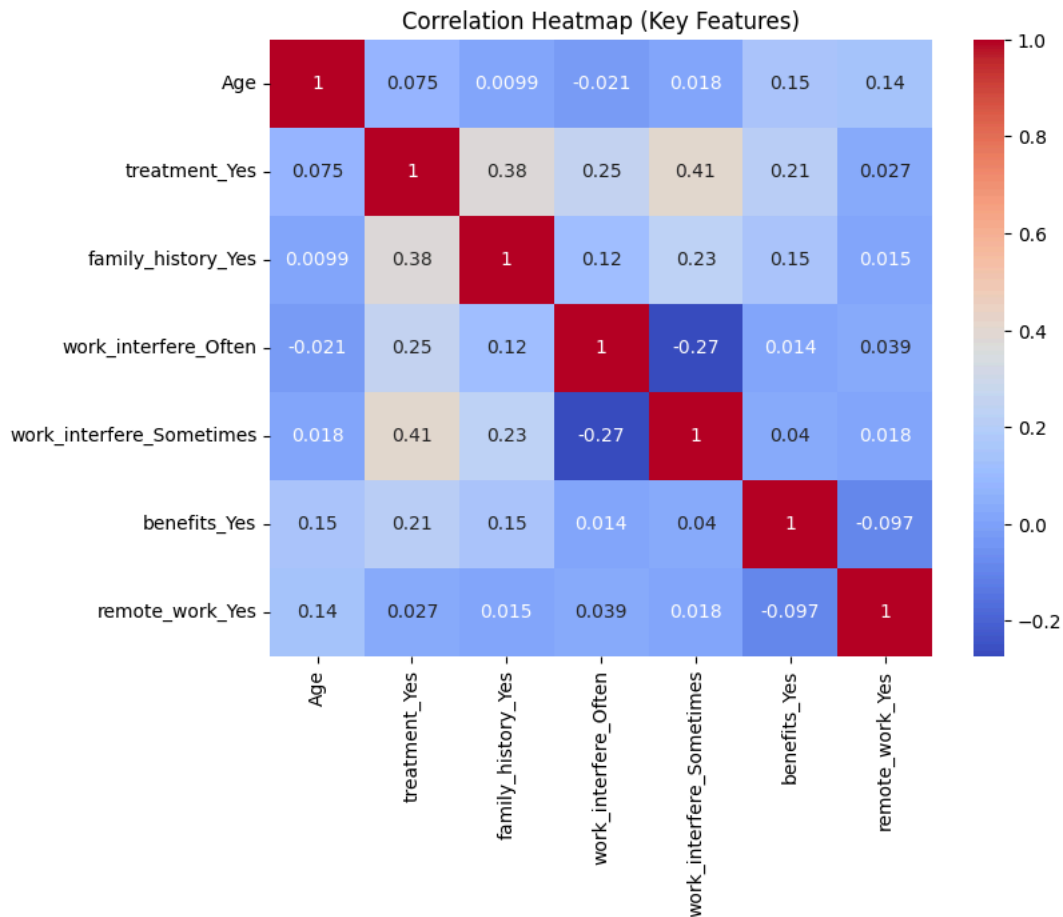
# Convert categorical variables using one-hot encoding
df_ml = pd.get_dummies(df_ml, drop_first=True)
```

Categorical variables were converted into numerical format using one-hot encoding, allowing machine learning models to process the data.

Correlation heatmap(only key features):


```
important_cols = [
    'Age',
    'treatment_Yes',
    'family_history_Yes',
    'work_interfere_Often',
    'work_interfere_Sometimes',
    'benefits_Yes',
    'remote_work_Yes'
]

plt.figure(figsize=(8,6))
sns.heatmap(df_ml[important_cols].corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap (Key Features)")
plt.show()
```



The correlation heatmap shows that family history and work interference have the strongest positive relationships with seeking mental health treatment. Workplace factors such as benefits also show some influence, while variables like remote work have minimal impact. These findings are consistent with the earlier statistical analysis and machine learning results.

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Splitting the Data:

```
X = df_ml.drop('treatment_Yes', axis=1)
y = df_ml['treatment_Yes']

from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```

The dataset was split into training (80%) and testing (20%) sets to evaluate the model's performance on unseen data.

Model Training (Logistic Regression):

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=5000, random_state=42)
model.fit(X_train, y_train)
```

▼ **LogisticRegression** ⓘ ?
LogisticRegression(max_iter=5000, random_state=42)

Logistic Regression was used as a baseline model for binary classification, as it is simple and interpretable.

Evaluating The Model:

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Model Accuracy:", round(accuracy * 100, 2), "%")
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Model Accuracy: 81.27 %

Confusion Matrix:

```
[[ 98 26]
 [ 21 106]]
```

Classification Report:

	precision	recall	f1-score	support
False	0.82	0.79	0.81	124
True	0.80	0.83	0.82	127
accuracy			0.81	251
macro avg	0.81	0.81	0.81	251
weighted avg	0.81	0.81	0.81	251

The model achieved an accuracy of approximately 81%, indicating that the selected features can reasonably predict whether an employee seeks treatment.

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Additional Model: Random Forest:

```
from sklearn.ensemble import RandomForestClassifier

rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
```

```
rf_pred = rf_model.predict(X_test)

from sklearn.metrics import accuracy_score

rf_accuracy = accuracy_score(y_test, rf_pred)

print("Random Forest Accuracy:", round(rf_accuracy * 100, 2), "%")
```

Random Forest Accuracy: 80.08 %

```
print("Logistic Regression Accuracy:", round(accuracy * 100, 2), "%")
print("Random Forest Accuracy:", round(rf_accuracy * 100, 2), "%")
```

Logistic Regression Accuracy: 81.27 %
Random Forest Accuracy: 80.08 %

A Random Forest model was implemented as a more complex alternative to capture non-linear patterns in the data.

Model Comparison(Interpretation):

Both Logistic Regression and Random Forest achieved similar accuracy (~80–82%), indicating that workplace and demographic factors can reasonably predict whether an employee seeks mental health treatment.

The similarity in performance suggests that:

- The relationships in the data are not highly complex
- Simpler models like Logistic Regression perform effectively
- More complex models like Random Forest do not provide a significant improvement

Additionally, feature importance and earlier statistical tests show that:

- Family history
- Work interference

are strong predictors of mental health treatment.

This confirms consistency between statistical analysis and machine learning results.

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Feature Importance (Model Interpretation):

```
coefficients = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.coef_[0]
}).sort_values(by="Importance", ascending=False)

coefficients.head(10).reset_index(drop=True)
```

	Feature	Importance
0	work_interfere_Sometimes	1.560648
1	work_interfere_Often	1.288739
2	work_interfere_Rarely	1.003744
3	benefits_Yes	0.597876
4	family_history_Yes	0.567708
5	Country_United States	0.481395
6	coworkers_Yes	0.439669
7	anonymity_Yes	0.403599
8	state_AZ	0.314715
9	care_options_Yes	0.311534

Next steps: [Generate code with coefficients](#) [New interactive sheet](#)

Feature importance analysis shows how each variable influences the prediction.

Work interference and family history were among the strongest predictors, confirming earlier statistical findings.

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✓ *Final Insights :*

✓ *Key Findings*

1. Work interference and family history emerged as the strongest predictors of mental health treatment, supported by both statistical tests and machine learning models.
2. Workplace conditions (such as support, benefits, and mental health impact on work) play a significant role in influencing treatment decisions.
3. The dataset shows that mental health concerns are relatively common in the tech industry, with a balanced distribution of treatment vs. non-treatment.
4. Machine learning models (Logistic Regression and Random Forest) achieved ~80–82% accuracy, indicating that employee behavior can be reasonably predicted using available features.

Limitations

- The data is self-reported, which may introduce response bias and affect reliability.
- The dataset is focused on tech industry employees, so results may not generalize to other industries.
- Some variables required imputation and cleaning, which may slightly impact accuracy.
- The models used are relatively simple and may not capture all complex real-world factors influencing mental health decisions.

Conclusion

This project demonstrates that both personal factors (like family history) and workplace conditions (like work interference and support) play a significant role in predicting mental health treatment behavior.

The combination of statistical analysis and machine learning provides consistent evidence that these factors are meaningful and can be used to build predictive models.

This project demonstrates how both statistical analysis and machine learning can be combined to better understand mental health trends